SQL-BASED SUPPLY CHAIN DATA ANALYSIS



SQL Server - Business Questions

1. Data exploration and preparation

01

How many records are present in the dataset?

```
SELECT COUNT(*) AS Total_Rows
FROM [dbo].[supply_chain_dataset];
```

	Total_Rows
1	180519

SQL Server - Business Questions

1. Data exploration and preparation

02

How many unique orders are present?

```
SELECT COUNT(DISTINCT [Order_Id]) AS Total_Orders
FROM dbo.supply_chain_dataset;
```

	Total_Orders
1	65752

SQL Server - Business Questions

1. Data exploration and preparation

03

Are the important columns stored in the correct data types?

```
SELECT
    COLUMN_NAME,
    DATA_TYPE,
    CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'supply_chain_dataset'
ORDER BY ORDINAL_POSITION;
```

	COLUMN_NAME	DATA_TYPE
1	Type	nvarchar
2	Days_for_shipping_real	tinyint
3	Days_for_shipment_scheduled	tinyint
4	Benefit_per_order	float
5	Sales_per_customer	float
6	Delivery_Status	nvarchar
7	Late_delivery_risk	bit
8	Category_Id	tinyint
9	Category_Name	nvarchar
10	Customer_City	nvarchar
11	Customer_Country	nvarchar
12	Customer_Email	nvarchar
13	Customer_Fname	nvarchar
14	Customer_Id	int
15	Customer_Lname	nvarchar
16	Customer_Password	nvarchar
17	Customer_Segment	nvarchar
18	Customer_State	nvarchar
19	Customer_Street	nvarchar
20	Customer_Zipcode	nvarchar
21	Department_Id	tinyint
22	Department_Name	nvarchar
23	Latitude	float
24	Longitude	float
25	Market	nvarchar
26	Order_City	nvarchar
27	Order_Country	nvarchar
28	Order_Customer_Id	int
29	order_date_DateOrders	datetime2
30	Order_Id	int
31	Order_Item_Cardprod_Id	int
32	Order_Item_Discount	float
33	Order_Item_Discount_Rate	float
34	Order_Item_Id	int
35	Order_Item_Product_Price	float
36	Order_Item_Profit_Ratio	float
37	Order_Item_Quantity	tinyint
38	Sales	float
39	Order_Item_Total	float
40	Order_Profit_Per_Order	float
41	Order_Region	nvarchar
42	Order_State	nvarchar
43	Order_Status	nvarchar
44	Order_Zipcode	nvarchar
45	Product_Card_Id	int
46	Product_Category_Id	tinyint
47	Product_Description	nvarchar
48	Product_Image	nvarchar
49	Product_Name	nvarchar
50	Product_Price	float
51	Product_Status	tinyint
52	shipping_date_DateOrders	datetime2
53	Shipping_Mode	nvarchar
54	Real Shipping Date	date
55	Order Date	date
56	Order Time	time
57	Real Shipping Time	time
58	Customer Full Name	varchar

SQL Server - Business Questions

1. Data exploration and preparation

04

How can shipping dates be calculated using order dates and shipping duration?

	Order_Id	OrderDate	Real Shipping Date	ShippingDuration	CalculatedShippingDuration	ValidationStatus
1	77202	2018-01-31 22:56:00.0000000	2018-02-03	3	3	Valid
2	75939	2018-01-13 12:27:00.0000000	2018-01-18	5	5	Valid
3	75938	2018-01-13 12:06:00.0000000	2018-01-17	4	4	Valid
4	75937	2018-01-13 11:45:00.0000000	2018-01-16	3	3	Valid
5	75936	2018-01-13 11:24:00.0000000	2018-01-15	2	2	Valid
6	75935	2018-01-13 11:03:00.0000000	2018-01-19	6	6	Valid
7	75934	2018-01-13 10:42:00.0000000	2018-01-15	2	2	Valid
8	75933	2018-01-13 10:21:00.0000000	2018-01-15	2	2	Valid
9	75932	2018-01-13 10:00:00.0000000	2018-01-16	3	3	Valid
10	75931	2018-01-13 09:39:00.0000000	2018-01-15	2	2	Valid

```sql
ALTER TABLE [dbo].[supply_chain_dataset]
ADD [Real Shipping Date] DATE;

UPDATE [dbo].[supply_chain_dataset]
SET [Real Shipping Date] =
    DATEADD(
        DAY,
        [Days_for_shipping_real],
        CAST([order_date_DateOrders] AS DATE)
    );
```

```sql
SELECT
    [Order_Id],
    [order_date_DateOrders] AS OrderDate,
    [Real Shipping Date],
    [Days_for_shipping_real]  AS ShippingDuration,
    DATEDIFF(
        DAY,
        [order_date_DateOrders],
        [Real Shipping Date]
    ) AS CalculatedShippingDuration,
    CASE
        WHEN [Real Shipping Date] IS NULL THEN 'Missing Shipping Date'
        WHEN [order_date_DateOrders] IS NULL THEN 'Missing Order Date'
        WHEN DATEDIFF(
            DAY,
            [order_date_DateOrders] ,
            [Real Shipping Date]
        ) = [Days_for_shipping_real]  THEN 'Valid'
        ELSE 'Mismatch'
    END AS ValidationStatus
FROM dbo.supply_chain_dataset;
```

SQL Server - Business Questions

2. Overall business performance

05

What are the total sales?

```sql
SELECT
    CONCAT('$', FORMAT(SUM(Sales) / 1000000.0, 'N2'), 'M') AS Total_Sales_Millions
FROM supply_chain_dataset;
```

	Total_Sales_Millions
1	$36.78M

SQL Server - Business Questions

2. Overall business performance

06

What is the total profit?

```sql
SELECT
    CONCAT('$', FORMAT(SUM([Order_Profit_Per_Order]) / 1000000.0, 'N2'), 'M')
        AS Total_Profit_M
FROM dbo.supply_chain_dataset;
```

	Total_Profit_M
1	$3.97M

SQL Server - Business Questions

2. Overall business performance

07

What is the overall profit margin?

```sql
SELECT
    ROUND(
        SUM([Order_Profit_Per_Order]) * 100.0
        / NULLIF(SUM([Sales]), 0),
        2
    ) AS Profit_Margin_Percent
FROM [dbo].[supply_chain_dataset];
```

	Profit_Margin_Percent
1	10.78

SQL Server - Business Questions

2. Overall business performance

08

What is the average sales value per order?

	Average_Sales_Per_Order
1	559.45

```
SELECT
    round(SUM([Sales]) / COUNT(DISTINCT [Order_Id]),2) AS Average_Sales_Per_Order
FROM dbo.supply_chain_dataset;
```

SQL Server - Business Questions

3. Product category analysis

09

Which product categories generate the highest sales?

```sql
SELECT top 1
    Category_Name,
    CONCAT('$', FORMAT(SUM(Sales) / 1000000.0, 'N2'), 'M') AS Total_Sales
FROM dbo.supply_chain_dataset
GROUP BY Category_Name
ORDER BY Total_Sales DESC;
```

	Category_Name	Total_Sales
1	Fishing	$6.93M
2	Cleats	$4.43M
3	Camping & Hiking	$4.12M
4	Cardio Equipment	$3.69M
5	Women's Apparel	$3.15M
6	Water Sports	$3.11M
7	Indoor/Outdoor Games	$2.89M
8	Men's Footwear	$2.89M
9	Shop By Sport	$1.31M
10	Computers	$0.66M

SQL Server - Business Questions

3. Product category analysis

10

Which product categories generate the highest profit?

```sql
SELECT top 1
    Category_Name,
    CONCAT('$', FORMAT(SUM([Order_Profit_Per_Order]) / 1000000.0, 'N2'), 'M') AS Total_Profit
FROM dbo.supply_chain_dataset
GROUP BY Category_Name
ORDER BY Total_Profit DESC;
```

	Category_Name	Total_Profit
1	Fishing	$0.76M
2	Cleats	$0.49M
3	Camping & Hiking	$0.43M
4	Cardio Equipment	$0.38M
5	Women's Apparel	$0.35M
6	Water Sports	$0.33M
7	Indoor/Outdoor Games	$0.32M
8	Men's Footwear	$0.31M
9	Shop By Sport	$0.13M
10	Computers	$0.07M

SQL Server - Business Questions

3. Product category analysis

11

Which product categories have the lowest profit margins?

```sql
SELECT top 10
    Category_Name,
    SUM(Sales) AS Total_Sales,
    SUM(Order_Profit_Per_Order) AS Total_Profit,
    CAST(
        SUM(Order_Profit_Per_Order) * 100.0 /
        NULLIF(SUM(Sales), 0)
        AS DECIMAL(10,2)
    ) AS Profit_Margin_Percentage
FROM dbo.supply_chain_dataset
GROUP BY Category_Name
ORDER BY Profit_Margin_Percentage ASC;
```

	Category_Name	Total_Sales	Total_Profit	Profit_Margin_Percentage
1	Strength Training	54895.5295410156	332.310089826584	0.61
2	As Seen on TV!	20597.9399642944	714.429979801178	3.47
3	Men's Clothing	43856.8012695313	2006.03996968269	4.57
4	Basketball	27099.3293457031	1845.67000293732	6.81
5	Books	12587.399969101	883.010007709265	7.02
6	Video Games	33310.5	2717.51999667287	8.16
7	DVDs	79395.5423583984	6655.43008828163	8.38
8	Pet Supplies	41524.8007507324	3589.25995850563	8.64
9	Health and Beauty	106080.483093262	9493.62997484207	8.95
10	Kids' Golf Clubs	98797.5792388916	9045.83989572525	9.16

SQL Server - Business Questions

4. Market analysis

12

Which markets generate the highest sales?

```sql
SELECT
    Market,
    CONCAT('$', FORMAT(SUM(Sales) / 1000000.0, 'N2'), 'M') AS Total_Sales
FROM dbo.supply_chain_dataset
GROUP BY Market
ORDER BY Total_Sales DESC;
```

	Market	Total_Sales
1	Pacific Asia	$8.27M
2	USCA	$5.07M
3	Africa	$2.29M
4	Europe	$10.87M
5	LATAM	$10.28M

SQL Server - Business Questions

4. Market analysis

13

Which markets generate the highest profit?

```sql
SELECT
    Market,
    CONCAT('$', FORMAT(SUM([Order_Profit_Per_Order]) / 1000000.0, 'N2'), 'M') AS Total_Profit
FROM dbo.supply_chain_dataset
GROUP BY Market
ORDER BY Total_Profit DESC;
```

	Market	Total_Profit
1	Europe	$1.17M
2	LATAM	$1.12M
3	Pacific Asia	$0.86M
4	USCA	$0.56M
5	Africa	$0.25M

SQL Server - Business Questions

4. Market analysis

14

How does profitability vary across different markets?

```
SELECT
    Market,
    SUM(Sales) AS Total_Sales,
    SUM(Order_Profit_Per_Order) AS Total_Profit,
    CAST(
        SUM(Order_Profit_Per_Order) * 100.0 /
        NULLIF(SUM(Sales), 0)
        AS DECIMAL(10,2)
    ) AS Profit_Margin_Percentage
FROM dbo.supply_chain_dataset
GROUP BY Market
ORDER BY Profit_Margin_Percentage DESC;
```

	Market	Total_Sales	Total_Profit	Profit_Margin_Percentage
1	USCA	5066528.7109108	564313.780372769	11.14
2	Africa	2294452.92517662	252071.180070952	10.99
3	LATAM	10277612.8408318	1123321.61147487	10.93
4	Europe	10872396.7955751	1169442.96097703	10.76
5	Pacific Asia	8273743.74056339	857753.441097483	10.37

SQL Server - Business Questions

5. Shipping and delivery analysis

15

How many records have a late-delivery status?

```sql
SELECT
    COUNT(*) AS Late_Delivery_Records
FROM dbo.supply_chain_dataset
WHERE Delivery_Status = 'Late delivery';
```

	Late_Delivery_Records
1	98977

SQL Server - Business Questions

5. Shipping and delivery analysis

16

What is the late-delivery percentage, excluding shipping-cancelled records?

```sql
SELECT
    COUNT(CASE
        WHEN Delivery_Status = 'Late delivery' THEN 1
    END) AS Late_Delivery_Records,

    COUNT(*) AS Total_Non_Cancelled_Records,

    CAST(
        COUNT(CASE
            WHEN Delivery_Status = 'Late delivery' THEN 1
        END) * 100.0 / NULLIF(COUNT(*), 0)
        AS DECIMAL(10,2)
    ) AS Late_Delivery_Percentage

FROM dbo.supply_chain_dataset
WHERE Delivery_Status <> 'Shipping canceled';
```

	Late_Delivery_Records	Total_Non_Cancelled_Records	Late_Delivery_Percentage
1	98977	172765	57.29

SQL Server - Business Questions

5. Shipping and delivery analysis

17

How does actual shipping duration compare with scheduled shipping duration?

```
SELECT
    round(AVG(CAST(DATEDIFF(DAY, order_date_DateOrders, [Real Shipping Date]) AS FLOAT)),2)
        AS Average_Actual_Shipping_Days,

    round(AVG(CAST(Days_for_shipping_real AS FLOAT)),2)
        AS Average_Recorded_Shipping_Days

FROM dbo.supply_chain_dataset
WHERE [Real Shipping Date] IS NOT NULL;
```

	Average_Actual_Shipping_Days	Average_Recorded_Shipping_Days
1	3.5	3.5

SQL Server - Business Questions

5. Shipping and delivery analysis

18

Which shipping methods have different shipping-performance patterns?

```sql
SELECT
    Shipping_Mode,
    COUNT(*) AS Total_Orders,

    COUNT(CASE
        WHEN Delivery_Status = 'Late delivery' THEN 1
    END) AS Late_Delivery_Records,

    CAST(
        COUNT(CASE
            WHEN Delivery_Status = 'Late delivery' THEN 1
        END) * 100.0 / NULLIF(COUNT(*), 0)
        AS DECIMAL(10,2)
    ) AS Late_Delivery_Percentage

FROM dbo.supply_chain_dataset
WHERE Delivery_Status <> 'Shipping canceled'
GROUP BY Shipping_Mode
ORDER BY Late_Delivery_Percentage DESC;
```

	Shipping_Mode	Total_Orders	Late_Delivery_Records	Late_Delivery_Percentage
1	First Class	26513	26513	100.00
2	Second Class	33806	26987	79.83
3	Same Day	9293	4454	47.93
4	Standard Class	103153	41023	39.77

THANK YOU

